

Consolidated Calculation Sheet

Every equation below is implemented and unit-tested; the bounded, deduplicated forms supersede the V6–V9 drafts.

✓ 36 / 36 checks pass

Scale: states in [0,1], C in (0,1), Q in [0,100]. Coefficients are prototype [P] pending calibration.

1 · Observation contribution

Each observation is scored, decayed by age, then bounded. Freshness halves every half-life H .

$$f_i = \exp(-\ln 2 \cdot \text{age} / H)$$

$$b_i = \text{severity} \cdot \text{confidence} \cdot f_i$$

Optional bounded modifiers (each neutral = 1) fold in capacity, redundancy, load and duplicate-damping:

$$c_i = b_i (0.25 + 0.75 \cdot \text{cap})(0.25 + 0.75(1 - \text{red}))(1 + \text{dev} + \text{ins})$$

Verified: zero severity $\rightarrow 0$; older < newer; stays bounded.

2 · Bounded domain state

Any pile of contributions maps into [0, 1). This removes the V6 unbounded-count defect: more reports can never push a domain past 1.

$$D_j = 1 - \exp(-\sum c_i)$$

Verified: $1 \times = 0.46$, $3 \times = 0.85$, $10 \times = 1.00$, $50 \times$ still < 1.

3 · Channel merge (dedupe)

Each driver takes the strongest of its sources instead of summing them, so a duplicated report is never counted twice.

$$E^* = \max(E, K_E, \text{EnergyLog}); T^* = \max(T, K_T, \text{Maritime, Delay})$$

$$I^* = \max(I, K_I, \text{Pipeline, Port}); G^* = \max(G, \text{Kinetic})$$

4 · Transmission sub-states

Sector states reuse the bounded form on their own contributions; Delay weights by deviation and capacity.

$$\text{Maritime, Pipeline, Port} = 1 - \exp(-\sum c_i)$$

$$\text{Delay} = 1 - \exp(-\sum b_i d_i \text{cap})$$

5 · Friction and condition index

Friction is linear in the domain states with pairwise interactions; the logistic squash keeps C in (0, 1) with no hard clip, so ranking never saturates to a flat 1 (the V6 defect).

$$F = b_0 + \sum w_j D_j + \sum \gamma_{jk} D_j D_k$$

$$C_t = 1 / (1 + \exp(-F))$$

Verified: C strictly in (0,1), monotone in F . Caveat: resolution compresses at the top (report F alongside C).

6 · Aggregate evidence confidence

A bundle mixes mean quality with the weakest item (minimum over all items), on a 0–100 scale.

$$Q = \text{clip}_{[0,100]} [0.75 \cdot \text{mean}(s) + 0.25 \cdot \min(s)]$$

Derived floor. Since $\text{mean} \leq 100$, to clear a gate at $\tau = 80$:

$$Q \leq 75 + 0.25 s_{\min} \Rightarrow s_{\min} \geq 4\tau - 300 = 20$$

Verified: $99 \times 100 + \text{one } 0 \rightarrow Q = 74.25$ (blocked). A zero-weakest bundle can never reach 80.

7 · Topology resilience (corrected)

The old form subtracted a redundancy term and could go negative. The corrected form attenuates criticality by redundancy, so the argument is never negative and R stays in [0, 1).

$$R = 1 - \exp(-\sum \text{crit}_i (1 - \text{red}_i))$$

Verified: old form returned $R = -1.01$ on a resilient node; corrected form stays in [0,1) and falls as redundancy rises.

8 · Prediction gate (no-trade band)

A forecast acts only outside a band set by the larger of expanding-window RMSE and a floor. Output is triage, not advice.

$$\hat{r}_{a,t+h} = \alpha + \beta^T Z_t$$

NEUTRAL if $|\hat{r}| \leq \max(\text{RMSE, threshold})$, else LONG / SHORT

9 · Macro mappings (estimable)

Linear-in-parameters, so OLS with Newey–West errors and an expanding window applies. θ is the expanding 90th-percentile shock flag (no look-ahead).

$$\Delta \pi = \beta_0 + \beta_1 E^* + \beta_2 T^* + \beta_3 M + \beta_4 G^* + \dots + \varphi \pi_t + \varepsilon$$

$$r_{a,t+h} = \alpha_a + B_a D_t + \lambda_a C_t + \varepsilon$$

Caveat: ~10 params on ~60 monthly obs is low-powered; E-interactions are collinear. Coefficients require real chronological calibration.

10 · Calibration metrics

Q is a triage score, not a probability. Before scoring, map to [0,1] with $p = Q/100$ (or calibrate). Then:

$$\text{Brier} = (1/n) \sum (p_i - y_i)^2$$

$$\text{ECE} = \sum_b (n_b/n) | \text{acc}(b) - \text{conf}(b) |$$

Brier decomposes into reliability + resolution – uncertainty. Verified: perfect=0, worst=1, and 95%25 confidence on 70%25-true events $\rightarrow \text{ECE} = 0.25$.